From Daily to Hourly Marine Heatwaves: A Taxonomy-Based, GPU-Accelerated Framework for Event Detection and Classification

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ABSTRACT

This study presents a methodology for detecting, analysing, and classifying Marine heatwaves (MHWs) at hourly resolution. MHWs are traditionally identified from daily Sea Surface Temperature (SST) records, yet this framework does not transfer directly to higher temporal resolution. At hourly scale, SST reveals sub-daily variability that is largely smoothed out in daily means, including intermittent threshold exceedances, short-lived interruptions, diurnal cycling, and nocturnal heat retention. These features complicate the definition of event continuity and persistence, but also provide an opportunity to examine the internal structure of marine heat extremes in substantially greater detail. Events are identified through a duration–gap framework that preserves short interruptions while maintaining event continuity, and are subsequently decomposed into individual diurnal cycles. Their intra-event morphology is used to construct an hourly-scale taxonomy that characterises MHWs not only in terms of conventional duration and intensity, but also through persistence, interruption structure, cycle-to-cycle evolution, diurnal heating, nocturnal retention, and other sub-daily dynamics that are not accessible from daily observations. To enable the methodology to scale from individual time series to multi-year gridded SST datasets, a GPU-accelerated implementation, `CUDAHeat`, is developed using CUDA-enabled array computation. The computational architecture preserves the complete temporal record of each spatial point while partitioning only the spatial domain into dynamically sized batches according to available GPU memory. VRAM-aware batch adaptation, automatic out-of-memory recovery, and explicit support for both 32-bit and 64-bit floating-point computation allow the same workflow to operate across different GPU memory capacities without altering the temporal definition of an event.
Benchmarking on a hourly gridded SST dataset comprising approximately 1.16 billion values shows a 23.4× acceleration of the core computation using GPU 32-bit arithmetic relative to the CPU 64-bit reference, with a 15.9× speedup for the complete processing pipeline. Scientific validation yields approximately 99.9991% agreement in threshold-exceedance classification across tens of millions of comparisons, with the remaining differences confined predominantly to values extremely close to the detection threshold. These results demonstrate that the hourly framework can retain its temporal and taxonomic detail while remaining computationally tractable for substantially larger spatial datasets. This combination of an hourly event taxonomy with scalable GPU computation provides a framework for investigating marine heat extremes from their conventional daily representation down to their sub-daily structure, and establishes a basis for subsequent analysis of coupled marine–atmospheric heat extremes and coastal thermal stress.

1. Introduction

1.1. State of the art

Some bibliographic results explain

1.2. Motivation

Why is this meaningful and useful

1.3. Data Used

Copernicus L4 gridded satellite data.

1.4. Github Repository

There is a dedicated GitHub Repository (at the moment still private and invisible), containing code and deployment instructions.

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2. Architecture

2.1. Crucial Features

2.1.1. Chunking along space

The computational domain is partitioned exclusively along its spatial dimensions, while the complete temporal sequence associated with each spatial point is preserved. This constraint ensures that event continuity, persistence, and sub-daily temporal structure remain available throughout the calculation.

2.1.2. Batching and Out of Memory (OOM)

Spatial batches are dynamically adapted according to the GPU memory available at execution time, allowing the same workflow to operate across devices with different VRAM capacities. If an out-of-memory condition occurs, the batch size is reduced and the affected spatial subset is retried without truncating or partitioning its temporal record.

2.1.3. Conversion to Float 32

Single-precision floating-point representation reduces both memory requirements and computational cost relative to double precision, allowing larger spatial batches to be processed on the GPU. The numerical impact of this conversion is explicitly evaluated against a double-precision CPU reference to quantify any differences introduced by reduced precision.

2.1.4. Preserved Consistency

GPU-derived results are systematically compared with the reference CPU implementation to verify that computational acceleration does not materially alter the scientific outcome. Consistency is evaluated for intermediate fields, threshold-exceedance classifications, and derived quantities, with particular attention to differences close to the detection threshold.

2.1.5. Improved Performance

GPU execution substantially reduces the computational time required to process large hourly SST datasets compared with the corresponding CPU implementation. Performance is quantified through execution time, speedup, throughput, memory consumption, and the relative contributions of host-to-device transfer, GPU computation, and device-to-host transfer.

3. Benchmark and Sensitivity Analysis

Describe the benchmark test, what how and why.

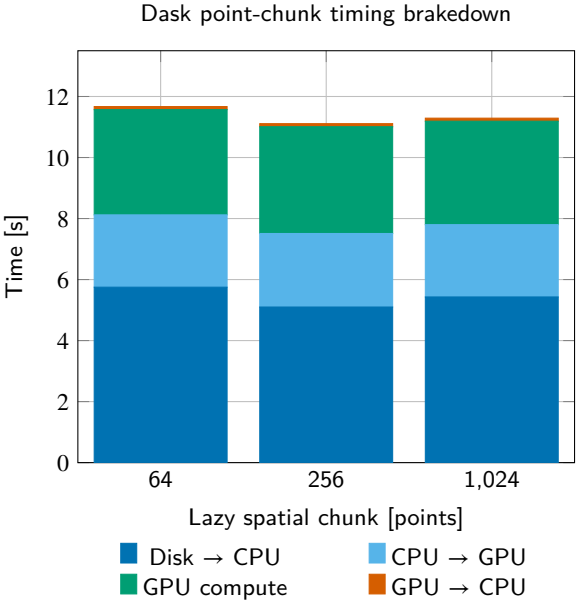


Figure 1: Dask point-chunk timing breakdown for the CUDAHeat full-pipeline benchmark, showing disk-to-CPU transfer, CPU-to-GPU transfer, GPU computation, and GPU-to-CPU transfer times for different spatial chunk sizes.

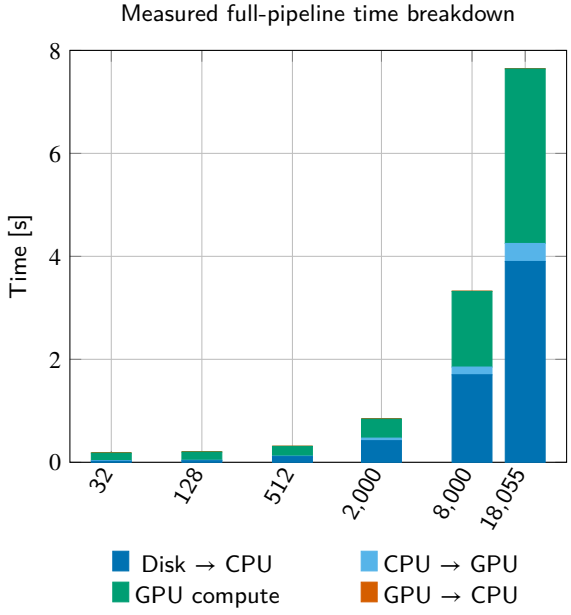


Figure 2: Measured CUDAHeat full-pipeline execution-time breakdown as a function of spatial dataset size, showing disk-to-CPU loading, CPU-to-GPU transfer, GPU computation, and GPU-to-CPU transfer times.

4. Methodology

4.1. Hourly Significance
4.2. Interanl Gaps
4.3. Joining sections

5. Taxonomy

5.1. Single-cycle
5.2. Total-cycle

6. Results

Some Results

7. Conclusions

Some conclusions

Some Appendix

References